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RNN, GRU, and LSTM Model Analysis for AI-Powered Indian Stock Market Index Forecasting

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Abstract*—The nonlinear and volatile nature of financial markets makes stock price prediction highly challenging. Traditional models, such as ARIMA, struggle to capture the complex relationships and unpredictable trends in stock market data. This research utilizes advanced deep learning methods, including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU), to predict stock prices based on historical data and key financial indicators. The YFinance library collects comprehensive financial data, while feature selection is enhanced through techniques such as hysteresis analysis, moving averages, and sentiment analysis using Natural Language Processing (NLP). Model performance is evaluated using metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared. Additionally, back-testing using historical data ensures the model's reliability. The results indicate that GRU outperforms other models in terms of computational efficiency and predictive accuracy. This study underscores the growing role of deep learning in financial analytics, providing investors with valuable insights and a structured, data-driven approach to decision-making in highly volatile market conditions.

***Keywords*—** *Stock Market Prediction, deep learning, RNN, GRU, LSTM.*

I. Introduction

The stock market is a fast-moving, unpredictable environment influenced by economic trends, political events, and investor emotions. Predicting stock prices has always been challenging because financial data doesn't follow a simple, linear pattern. Prices fluctuate suddenly due to global events, corporate earnings reports, and shifting market sentiment. Traditional forecasting methods, including statistical and econometric models, often struggle to capture these complexities, making them less effective in providing accurate predictions. However, with advancements in artificial intelligence and deep learning, stock market forecasting has evolved significantly. Deep learning models, particularly RNNs and their more advanced versions like LSTM and GRU, have shown great potential in analyzing financial time series. These models recognize sequential patterns, detect hidden trends, and learn from past stock movements, making them highly suitable for stock price prediction.

A. Forecasting stock prices is challenging for several reasons

High Volatility & Nonlinearity: Stock prices don't follow a predictable pattern. They are affected by a wide range of unpredictable factors, making traditional models, which assume stability and linearity, less effective. Influence of Multiple Factors Market trends depend on historical prices and trading volume, financial ratios, and economic indicators. External events such as political decisions, breaking news, and investor sentiment—can significantly impact stock prices. Limitations of Traditional Models: Conventional methods like Autoregressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS) are based on predefined mathematical assumptions. While they work well for stable time series, they struggle with capturing long-term dependencies and sudden market shifts. Data Inconsistencies & Noise: Financial data is often noisy, with missing values, outliers, and sudden fluctuations caused by market shocks. Filtering out irrelevant information while extracting meaningful insights is crucial for accurate forecasting.

B. Deep Learning for Stock Market Prediction

RNNs are designed to process sequential data. RNNs retain information from previous inputs, helping them analyze trends over time. However, they struggle with remembering long-term patterns due to the vanishing gradient problem. LSTM networks improve upon RNNs by introducing memory cells and gate mechanisms, allowing them to retain important information for longer periods. This makes them effective for stock price prediction. GRU simplifies LSTM structures while maintaining high performance. They use update and reset gates to manage information flow efficiently, making them computationally faster and more effective in financial forecasting.

C. Proposed Approach

This research applies deep learning techniques to predict stock prices using historical data from Yahoo Finance (YFinance). The dataset includes stock prices from major technology and financial companies, includingTechnology Sector: NVIDIA (NVDA), Microsoft (MSFT), Google (GOOGL), Apple (AAPL), Amazon (AMZN), Adobe (ADBE), Intel (INTC), Tesla (TSLA), IBM, and Salesforce (CRM). Financial Sector: PayPal (PYPL) and Square (SQ). OHLCV Data: The Open, High, Low, Close, and Volume (OHLCV) values, which form the foundation of stock market analysis. Moving Averages Both Simple Moving Average (SMA) and Exponential Moving Average (EMA) help capture long-term and short-term trends. Hysteresis Analysis: A method used to analyze past market behavior and identify repeating patterns. Sentiment Analysis using NLP to analyze the data. Financial news and social media trends to gauge

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investor sentiment and its potential impact on stock movements.

D. Evaluation Metrics and Model Comparison

MSE measures the average squared difference between predicted and actual values, penalizing larger errors. RMSE is the square root of MSE, providing a more interpretable measure of prediction accuracy. The Coefficient of Determination (R^2) evaluates how well the model explains the variance in stock prices, with values closer to 1 indicating better predictive performance. Through comparative analysis, this research aims to identify the most suitable deep learning model for stock price forecasting. Preliminary results suggest that the GRU outperforms both RNNs and LSTM models regarding computational efficiency and accuracy, making it a promising tool for financial market forecasting.

II. LITERATURE SURVEY

The study uses historical data to compare three deep learning models—LSTM, RNN, and GRU—for predicting stock prices. It addresses challenges such as non-linearity, volatility, and noise through preprocessing.The models are evaluated using metrics such as MSE, RMSE, and MAE. While GRU is more efficient, LSTM captures long-term dependencies better. The study suggests that deep learning models provide valuable insights for stock market forecasting. [1]

The paper proposes a hybrid ARIMA-LSTM model that combines linear trend capture (ARIMA) with non-linear residual handling (LSTM) for stock price prediction. It significantly outperforms the Prophet library, achieving much lower MSE and RMSE with a 99% fit. The model enhances prediction accuracy by integrating machine learning and deep learning techniques [2]

The paper introduces a deep learning framework combining a Bidirectional Gated Recurrent Unit (BiGRU) with an external activation layer for stock price prediction. The BiGRU model outperforms the Bidirectional Long Short Term Memory (BiLSTM) model in accuracy, stability, and efficiency, excelling in forecasting long-term prices and sudden price spikes. It offers a reliable, high-accuracy solution for informed trading decisions using NIFTY-50 data. [3]

The paper presents StockNet, a GRU-based model for stock index prediction, incorporating an Injection module to prevent overfitting and an Investigation module for forecasting. Validated on the CNX-Nifty index, StockNet outperforms traditional models by capturing both linear and non-linear trends, enhancing prediction accuracy. The model uses data augmentation and a modular design to improve robustness, trained with historical stock prices, trading volumes, and financial indicators. [4]

The paper investigates using LSTM models for predicting stock prices in the Indian share market, focusing on historical data and net growth calculations. It highlights challenges like investor sentiment and macroeconomic factors, demonstrating that LSTMs can capture temporal patterns for accurate predictions. The study suggests incorporating additional parameters like sentiment and geopolitical stability to further enhance prediction accuracy.[5]

The paper explores using LSTM models for stock price prediction in the Indian share market, demonstrating their ability to capture temporal patterns for accurate predictions. It suggests incorporating factors like investor sentiment and geopolitical stability to enhance prediction accuracy [6]

The paper presents an LSTM model for predicting Nifty50 stock prices, achieving 83.88% accuracy using ten years of historical data. It highlights the importance of preprocessing techniques and shows that LSTM effectively captures temporal patterns, offering valuable insights for investors.[7]

The paper compares deep learning models for stock price prediction in the Indian market, finding CNN and GRU to be the most effective, with improvements possible through feature engineering and hyperparameter optimization.[8]

The paper presents a hybrid LSTM-GRU model that outperforms others in predicting stock market indices, emphasizing the importance of preprocessing techniques for handling volatility and non-linearity.[9]

The paper presents a hybrid Recurrent Neural Network (HyRNN) model combining Bi-LSTM, GRU, and LSTM to predict stock prices by integrating stock features with financial news sentiments. It shows that incorporating sentiment analysis improves prediction accuracy and captures both linear and non-linear trends in stock market data.[10]

The paper "GRUvader" explores a sentiment-informed GRU model for stock price prediction, integrating lexicon-based sentiment features. It compares multiple ML algorithms, showing that GRUvader improves accuracy and stability. The study highlights sentiment analysis's role in capturing investor sentiment and external events using historical stock and Twitter sentiment data.[11]

The paper proposes an Attention GRU-XGBoost model for stock market prediction, integrating deep learning and boosting techniques to improve accuracy. It addresses volatility and trend prediction challenges, demonstrating better RMSE results than traditional models. The study highlights the importance of combining GRU, attention mechanisms, and XGBoost to capture both linear and non-linear trends effectively.[12].

TABLE I. SUMMARY OF LITERATURE SURVEY

Existing Algorithm / Methods / Framework/ Other Technical Details	***Findings / Result / Accuracy***	***Factors Observed / Factors to be Improved***
LSTM , RNN – GRU - Time Series Analysis - Financial Markets[1]	LSTM and GRU models generally showed better performance in capturing long-term dependencies and adapting to market changes - Performance was evaluated using MSE, RMSE, and MAE metrics	Each model has its strengths and weaknesses Potential for further improvement in handling more complex datasets.
ARIMA LSTM Hybrid Model - Time Series Forecasting - finance API -	The integrated ARIMA-LSTM approach	ARIMA-LSTM hybrid model

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Existing Algorithm / Methods / Framework/ Other Technical Details	***Findings / Result / Accuracy***	***Factors Observed / Factors to be Improved***
Prophet Forecasting Library [2]	outperformed other models, achieving an MSE of 3.03 and an RMSE of 1.74, which signifies a 99% model fit. On the other hand, the Prophet model showed less precision, resulting in a higher RMSE of 27.59 and MSE of 761.33	outperformed Prophet - Captures both linear and non-linear components of time series data - Potential for further improvement in handling more complex datasets
BiGRU, BiLSTM Time-Series Analysis – RNN NIFTY-50 Index[3]	The BiGRU model outperforms BiLSTM with better stability, smaller trainable parameters, and high accuracy in forecasting stock prices, including sudden spikes, up to 1000 days ahead.	BiGRU model demonstrated better stability and fewer parameters - Higher accuracy in forecasting stock prices - Potential for further improvement in handling more complex datasets.
(GRU) - Stock Market Index - Overfitting - Data Augmentation - CNX-Nifty - Injection Module - Investigation Module [4]	StockNet-model produced 65.59%, 27.30%, and 14.89% less test loss in terms of RMSE, MAE, and MAPE, respectively, compared to the TargetNet model - Improved prediction accuracy by reducing overfitting.	The injection module helps to prohibit overfitting by injecting diverse features - The investigation module focuses on stock index forecasting and the potential for further improvement in handling more complex datasets.
Long Short-Term Memory (LSTM) - Historical stock price data from the Indian share market, including companies like TCS, obtained from Yahoo Finance [5]	The LSTM model effectively predicted stock prices with high accuracy - Performance was evaluated using metrics such as MSE and Mean Absolute Percentage Error (MAPE)	The study emphasizes evaluating machine learning models across market conditions and incorporating external data, like macroeconomic indicators and market news, to enhance prediction accuracy.
RNN, LSTM, GRU-BiRNN - Time Series Analysis[6]	The BiRNN model outperforms Simple RNN, LSTM, and GRU with an MSE of 17740.79, MAE of 90.27, and MAPE of 0.81%.	The BiRNN model outperforms other RNN models in predicting stock prices for the Indian market's Nifty 50 index and highlights its potential for future research.
Deep Learning ,LSTM [7]	The model has strong predictive performance, with an average accuracy of 83% for the selected Nifty50 stocks and the highest accuracy for SBI at 83.88%	the Nifty50 stocks have the predicted errors, with SBI stock achieving an accuracy of 83.88%, and all models improved in forecast accuracy.
(RNN)(LSTM) -(CNN) (GRU) - Attention LSTM [8]	RNN, LSTM, CNN, GRU, and Attention LSTM are deep learning architectures used for sequence prediction and feature extraction.	Deep learning models across market conditions and incorporating external data like macroeconomic indicators and market news to enhance prediction performance.
(GRU) (LSTM) Ensemble Models Time Series Analysis Nike stock price data from January 2006 to January 2018[9]	The Hybrid LSTM-GRU model outperformed others, improving accuracy speed and addressing the vanishing gradient problem.	Potential for further improvement in handling more complex datasets - Need for more extensive evaluation of different stock indices.
RNN, GRU, LSTM Sentiment Analysis - Financial News - Time Series Analysis [10]	The hybrid RNN model showed significant improvement in prediction accuracy compared to traditional models - It effectively captured both stock features	Combining stock features with sentiment analysis of financial news improves prediction accuracy - Potential for further improvement in handling

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Existing Algorithm / Methods / Framework/ Other Technical Details	***Findings / Result / Accuracy***	***Factors Observed / Factors to be Improved***
	and news sentiments.	more complex datasets.
GRUvader, ARIMA, LSTM, Bi-LSTM, Bi-GRU, GAN, VADER Lexicon[11]	GRUvader outperformed other models with an accuracy of 97.60% for TSLA, 96.86% for AMD, 97.30% for META, and 96.14% for NIO.	Incorporating sentiment analysis significantly improves prediction performance, especially with GRUvader. GAN models showed poor performance with sentiment features.
GRU, XGBoost, Attention Mechanism, ARIMA, CNN, LSTM[12]	The GRU-XGBoost model attention demonstrated superior performance with better RMSE metrics compared to ARIMA and LSTM algorithms.	The model effectively handles heterogeneous data and improves prediction accuracy. The attention mechanism enhances the model's performance by adjusting the weight relationship between the current output and input.

III. METHODOLOGY

Proposed Framework: Enhancing Stock Price Prediction with Dynamic Deep Learning Models as Overview of the Proposed Model/Algorithm: The proposed model leverages the adaptive capabilities of LSTM, the sequential memory of RNN, and the streamlined structure of GRU.

A. RNNs:

RNNs are deep learning models designed for sequential data processing, making them effective for tasks like time series forecasting, speech recognition, and language translation. They use hidden states to capture dependencies between time steps, updating based on the current input and past states. Training is performed using Backpropagation Through Time (BPTT) and optimization techniques like gradient descent to minimize errors, often measured by MSE. However, RNNs face challenges like the vanishing gradient problem, which makes learning long-term dependencies difficult. Variants like LSTMs and GRUs help address these limitations for better performance.

B. LSTM:

LSTM networks are a type of RNN designed to overcome the vanishing gradient problem, enabling effective learning of long-term dependencies. They are widely used in language modeling, time series forecasting, and sequence generation. LSTMs use a cell state for memory retention and gates to regulate information flow: the forget gate discards unnecessary data, the input gate updates memory, and the output gate controls predictions. Training involves forward propagation, loss calculation, and BPTT with optimizers like Adam. Activation functions like tanh and sigmoid help regulate learning, making LSTMs ideal for sequential data processing.

C. GRUs

GRUs are a simplified variant of LSTMs designed to handle the vanishing gradient problem while capturing long-term dependencies. Used in language modeling, speech recognition, and time series forecasting, GRUs regulate information through update and reset gates. The update gate retains essential past information, while the reset gate discards unnecessary data. The candidate activation generates new information, updating the hidden state for context retention. GRUs train faster than LSTMs due to fewer parameters, efficiently managing memory while maintaining accuracy. Though less interpretable than simple RNNs, they offer a strong balance between performance and computational efficiency.

D. Collect the Live Stock Market Data

One of the primary challenges in this endeavor is obtaining real-time data while ensuring high accuracy. Various stock market data APIs offer up-to-the-minute information on financial instruments currently traded in the market. These interfaces enable access to both historical records and current prices of publicly listed stocks. Additionally, they help generate crucial indicators necessary for market surveillance and the development of trading tactics. As of 2019, some of the newest APIs include World Trading Data, Google Finance, and Yahoo's YFinance API. This project utilizes Yahoo Finance's API to collect real-time stock market information in OHLCV format

E. Data Preprocessing

Analyze stock market trends and forecast the Indian Stock Market Index using AI techniques, historical market data needs to be collected and processed. The history function retrieves this data, specifying parameters such as period, interval, start and end dates, and corporate actions like dividends and splits EDA plays a crucial role in understanding the impact of various features on the stock's closing price, which is a key indicator of market performance. One essential aspect of EDA is examining the closing price over time, as it represents the last traded price of a stock during a trading session and serves as a benchmark for investors. Moving averages (MAs) are widely used to identify trends by smoothing out short-term fluctuations. The SMA calculates the mean of closing prices over a set period, helping to highlight support and resistance levels while reducing price volatility. However, since SMA gives equal weight to all prices, it acts as a lagging indicator. In contrast, the EMA assigns greater weight to recent prices using a smoothing factor, making it more responsive to price changes and ideal for short-term trend analysis. While EMA is useful for identifying buy and sell signals quickly, it is also more prone to false signals in volatile markets. The dataset for

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model training typically spans from January 1, 2010, to the present, with a training-testing split: historical data up to one year before the present and the most recent year as the set. Handling missing values is crucial in ensuring data quality, and techniques like forward fill, backward fill, or interpolation can be used to fill gaps. Feature scaling, using Min-Max scaling or standardization, ensures that variables are on a similar scale, improving deep learning model performance. Data transformation is necessary for models like RNNs, LSTMs, and GRUs, where past sequence data are used to predict future trends. This involves defining a look-back period and creating sliding windows of sequential data points. The train-test split must maintain the chronological order of data to preserve the temporal structure, ensuring that training data precedes the test set for accurate model evaluation.

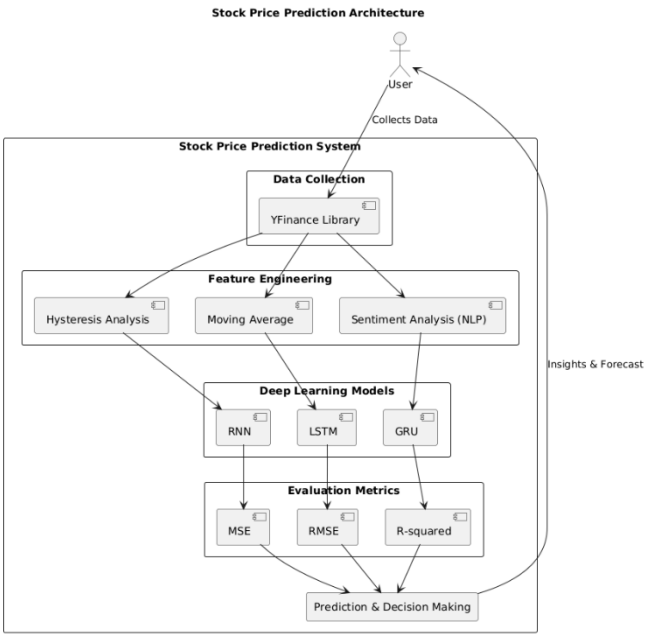

fig -1 Architecture Diagram

IV. Result & Discussion

Library and Python Keras . Our proposed model, after proper hyperparameter tuning and backtesting, was trained and validated using stock prices from the YFinance API dataset with an 80:20 training and testing data split. We tuned a GRU model, which serves as the baseline model in this work. A comparison between RNNs, LSTM, and GRU was conducted in terms of different evaluation metrics with changing hidden layers for data from NVDA, CRWD, AMD, ADBE, MSFT, GOOGL, INTC, AAPL, AMZN, CRM, IBM, TSLA, PYPL, and SQ. To optimize performance, we experimented with different look-back values to find the best fit. A short look-back might miss important patterns, leading to underfitting, while a long look-back may include too much noise, causing overfitting. Balance is key to capturing useful trends without unnecessary details. The performance of the three deep learning models—RNN (with tanh activation), LSTM, and GRU—was assessed on both the training and test datasets. The evaluation metrics include Mean Absolute Error (MAE), MSE , RMSE , and R^2 . The evaluation metrics provide a comprehensive view of the model's accuracy, error magnitude, and ability to generalize. All three models performed quite well during training, with R2 values approaching 1.0, suggesting an almost flawless fit to the training set.

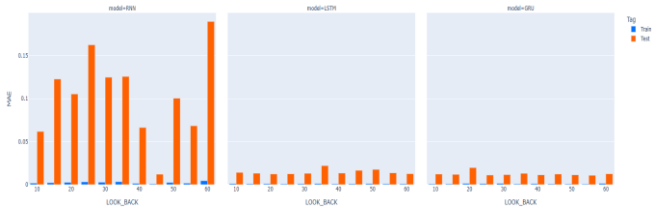

Fig.2 Model performance comparison: MAE (Train and Test)

The LSTM model had a training MAE of 0.001924, which, although slightly higher than RNN_ tanh, still indicated high accuracy. On the test data, LSTM achieved an MAE of 0.016611, lower than RNN_ tanh, highlighting its improved performance on unseen data. The GRU model demonstrated the lowest training MAE of 0.001813, confirming its high accuracy, and further excelled on the test data with an MAE of 0.012746, the lowest among all models, proving its superior generalization ability. In contrast, RNN_ tanh had the smallest training MAE of 0.001401, indicating excellent accuracy during training, but its test MAE increased to 0.027192, showing a significant drop in accuracy when applied to new data.

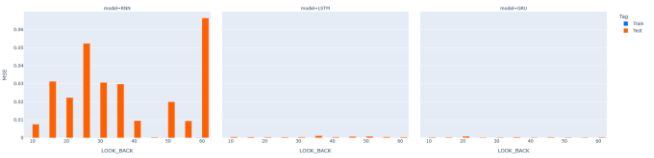

Fig.3 Model performance comparison: MSE (Train and Test)

The RNN_tanh model has a training Mean Squared Error (MSE) of 0.000005, indicating that the predictions are very close to the actual values with minimal squared errors. However, for the testing data, the MSE is 0.002240, which is higher and suggests that the model's precision decreases when applied to new, unseen data. The LSTM model has a training MSE of 0.000005, which is similar to the RNN_tanh model and indicates good accuracy in predictions. For the testing data, the LSTM model has an MSE of 0.000809, showing that its predictions are closer to the actual values compared to those of the RNN_tanh model. The GRU model has a training Mean Squared Error (MSE) of 0.000004, which is the lowest among the models, meaning its predictions are extremely close to the actual values. For the testing data, the GRU model also achieves the lowest MSE of 0.000437, indicating the best prediction performance on new, unseen data.

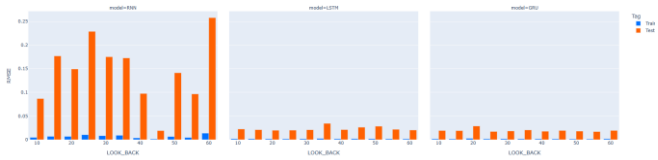

Fig -4 Model performance comparison RMSE(Train and Test)

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The RNN_tanh model has a training Root Mean Squared Error (RMSE) of 0.002124, derived from the MSE, which suggests high accuracy in predictions. However, its testing RMSE is 0.047326, indicating lower prediction accuracy on new data. On the other hand, the LSTM model's training RMSE is 0.002292, slightly higher but still within the range of accurate predictions. Its testing RMSE is 0.028439, which suggests higher accuracy on test data compared to RNN_tanh. The GRU model, showing the best performance, has a training RMSE of 0.002108, the lowest among the three models, indicating the best prediction accuracy. Its testing RMSE is 0.020900, confirming its superior accuracy on unseen data.

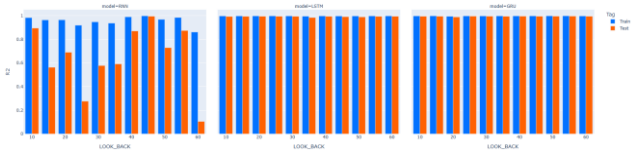

Fig -5 Model performance comparison: R2 (Train and Test)

The GRU model outperforms RNN and LSTM in stock price forecasting, achieving the highest R^2 values (99.669% training, 99.412% testing) and the lowest RMSE and MSE, ensuring superior accuracy and generalization. While RNN (99.664% training, 96.984% testing) and LSTM (99.608% training, 98.911% testing) also perform well, GRU excels in balancing precision, computational efficiency, and handling temporal dependencies. Its ability to maintain high accuracy on unseen test data makes it the most reliable model for stock price forecasting. Back-testing, a crucial step in model evaluation, was applied to validate these models using historical stock market data. The results show that all three models effectively capture historical trends, with GRU demonstrating the best generalization and predictive performance. The RNN model with tanh activation and a 60-look-back window achieved an R^2 of 99.664% on the training set and 96.984% on the test set, indicating slight generalization loss. LSTM performed better, with an R^2 of 99.608% on training and 98.911% on testing, showing strong predictive performance. However, GRU achieved the highest R^2 values, proving its superior ability to align predicted stock prices with actual trends.

The GRU model's robustness, efficiency, and ability to capture market trends make it the best choice for time-series forecasting, particularly for stock price prediction. Its effective generalization ensures reliability in real-world applications. Overall, back-testing confirms that GRU provides the most precise and stable forecasts, making it the ideal model for future stock price predictions.

model_name	MAE	MSE	RMSE	R2	n_samples	tag
RNN_tanh	0.001401	0.000005	0.002124	0.996637	2939	Train
RNN_tanh	0.027192	0.002240	0.047326	0.969837	690	Test
LSTM	0.001924	0.000005	0.002292	0.996084	2939	Train
LSTM	0.016611	0.000809	0.028439	0.989108	690	Test
GRU	0.001813	0.000004	0.002108	0.996687	2939	Train
GRU	0.012746	0.000437	0.020900	0.994117	690	Test

Fig-6 Comparison of models

The table compares the performance of RNN, LSTM, and GRU on training and test datasets. GRU demonstrates the best accuracy with the lowest error, making it the most effective model. LSTM performs better than RNN but is slightly less efficient than GRU. RNN struggles with generalization, showing higher errors on the test set. Overall, GRU is the preferred choice due to its superior performance and stability.

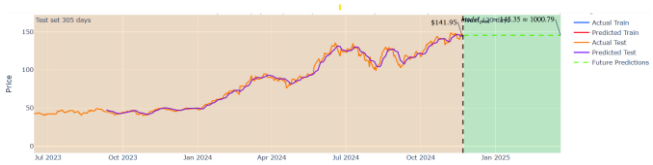

Fig –7 (NVDA) stock prices

Future prediction and backtesting This is a visual representation of a data set that is used to forecast the future using GRU. In Figure- 7, we employed a GRU model with a 60-day look-back period to forecast NVIDIA's (NVDA) stock prices. The dataset was divided into an 80:20 ratio for training and testing. The training set comprised 3,324 samples, while the testing set included data from November 23, 2024, to March 22, 2025, covering 120 days. The GRU model demonstrated strong predictive performance in forecasting NVIDIA's stock prices. In the training phase, it achieved an MAE of 0.0021, an MSE of 1.24e-05, a Root RMSE of 0.0035, and a Coefficient of Determination R^2 of 99.56%. During testing, the model recorded an MAE of 0.0207, an MSE of 8.28e-04, an RMSE of 0.0288, and an R^2 of 98.45%. These high R^2 values in both phases indicate that the GRU model effectively captures the underlying patterns in NVIDIA's stock prices, demonstrating strong predictive performance and generalization to future data.

CONCLUSION

The need for reliable and accurate financial market forecasting tools is rapidly increasing among businesses. This study shows that systems based on deep learning may successfully detect long-term dependencies in real-time data. To forecast the next 120 days, we compared RNN, LSTM, and Gru with real-time data from Yfinance data NVDA, CRWD, AMD, ADBE, MSFT, GOOGL, INTC, AAPL, AMZN, CRM, IBM, TSLA, PYPL, SQ data sets. According to the result analysis, GRU seems to be the most accurate model on both training and testing datasets, with the highest R2 and the lowest error metrics. It also performs well on the test data as well. Following closely behind, LSTM performs admirably but has somewhat higher error values on the test set than GRU. With the largest discrepancy between training and test outcomes, RNN performs the worst, suggesting that it is not as generalizable. For long-term forecasting, we aim to develop a more reliable prediction model capable of analyzing large datasets with numerous real-time data points goal is to enhance the LSTM RNN model's limitations.

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